

Total No. of Questions : 8]

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Roll No

CS-3005-CBGS

B.E. III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Discrete Structures

Time : Three Hours

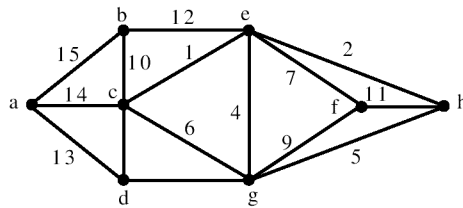
Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Among integers 1 to 300, how many of them are divisible neither by 3, nor by 5, nor by 7? How many of them are divisible by 3 but not by 5, nor by 7?
b) Prove by mathematical induction:
 $1 + 4 + 7 + \dots + (3n - 2) = 2n(3n - 1)$
2. a) Prove :
i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$
ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
b) Explain Pigeonhole principle with an example.
3. a) Show that the mapping $f: R \rightarrow R, f(x) = \frac{1}{x}, x \neq 0$ and $x \in R$ is one-one onto, where R is the set of non-zero real numbers.
b) Prove that every finite group G is isomorphic to a permutation group.

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4. a) Show that
$$\left[(p \wedge q) \Rightarrow p \right] \Rightarrow (q \wedge \sim q)$$
 is a contradiction.
b) Show that the language
$$L = \{a^m : m = i^2, i \geq 1\}$$
 is not a finite state
5. a) Prove by truth table that the following formula are tautologies
i) $(\sim q \Rightarrow \sim p) \wedge (q \Rightarrow p) \Rightarrow (p \Leftrightarrow q)$
ii) $(p \Leftrightarrow q \wedge r) \Rightarrow (\sim r \Rightarrow \sim p)$
b) Explain Pigeonhole principle with an example.
6. a) Find the shortest path of the following graph.



- b) Define chromatic number with diagram.
7. a) Solve the recurrence relation
$$a_{n+2} - 6a_{n+1} + 9a_n = 3(2^n) + 7(3^n), n \geq 0$$

given test $a_0 = 1$ and $a_1 = 4$.
b) Explain Binomial theorem and prove it. Take any one example to illustrate the theorem.
8. Write short notes:
a) Properties of Lattice
b) Multinomial coefficients
c) Graph coloring
d) Universal and existential quantitation
